Worksheet 3.3 — The Sine Rule and Cosine Rule

Mathematics: Analysis & Approaches SL (IB) · Unit 3: Geometry and Trigonometry

The Big Ideas

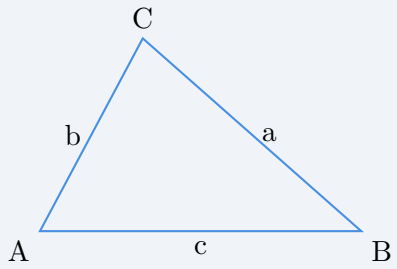
The sine and cosine rules solve any triangle, including the tricky ambiguous case.

- $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$; $a^2 = b^2 + c^2 - 2bc\cos A$.
- Area $= \frac{1}{2}ab\sin C$.
- Ambiguous case (SSA): check whether both θ and $180° - \theta$ give valid triangles.

Worked Examples

Example 1 — Sine rule for a side

In $\triangle ABC$, $A = 35°$, $B = 70°$, $a = 14$. Find b.



Solution. $\dfrac{14}{\sin 35°} = \dfrac{b}{\sin 70°} \Rightarrow b = \dfrac{14\sin 70°}{\sin 35°}$.
Conclusion: $b \approx 22.9$.

Example 2 — Cosine rule for the largest angle

A triangle has sides $6, 8, 11$. Find the largest angle.

Solution. Opposite 11: $\cos C = \dfrac{36 + 64 - 121}{96} = \dfrac{-21}{96}$.
Conclusion: $C \approx 102.6°$.

Example 3 — Ambiguous case

In $\triangle ABC$, $A = 40°$, $a = 9$, $b = 13$. Find both possible values of B.

Solution. $\sin B = \dfrac{13\sin 40°}{9} \approx 0.928$. $B_1 \approx 68.2°$; $B_2 = 180° - 68.2° = 111.8°$; check $A + B_2 = 151.8° < 180°$, valid.
Conclusion: $B \approx 68.2°$ or $111.8°$.

Example 4 — Navigation with the cosine rule

A ship sails 30 km on a bearing of 050°, then 45 km on a bearing of 140°. Find the direct distance home.

Solution. Included angle $= 140 - 50 = 90°$. $d^2 = 900 + 2025 - 0 = 2925$.
Conclusion: $d \approx 54.1$ km.

Example 5 — Cosine rule for a side, obtuse angle

In $\triangle ABC$, $b = 10$, $c = 14$, $A = 110$ř. Find a.

Solution. $a^2 = 100 + 196 - 2(10)(14)\cos 110$ř $= 296 - 280(-0.342)$.

Conclusion: $a^2 \approx 391.8 \Rightarrow a \approx 19.79$.

Example 6 — Ambiguous case with only one valid triangle

In $\triangle ABC$, $A = 110$ř, $a = 20$, $b = 15$. Find B, and explain why only one triangle is possible.

Solution. $\sin B = \dfrac{15 \sin 110$ř$}{20} \approx 0.7048 \Rightarrow B_1 \approx 44.8$ř. The supplementary option $B_2 = 135.2$ř would make $A + B_2 > 180$ř, impossible.

Conclusion: only $B \approx 44.8$ř is valid — an obtuse given angle always rules out the second case.

Example 7 — Area via the cosine rule first

Find the area of a triangle with sides $8, 10, 13$.

Solution. Find the angle opposite 13: $\cos C = \dfrac{64 + 100 - 169}{160} = -0.03125 \Rightarrow C \approx 91.8$ř.

Conclusion: Area $= \frac{1}{2}(8)(10)\sin(91.8$ř$) \approx 39.98$ square units.

Example 8 — Two-leg navigation distance

A plane flies 40 km on a bearing of 015ř, then 55 km on a bearing of 100ř. Find the distance between the start and end points.

Solution. Included angle $= 100 - 15 = 85$ř. $d^2 = 1600 + 3025 - 2(40)(55)\cos 85$ř.

Conclusion: $d^2 \approx 4241.3 \Rightarrow d \approx 65.1$ km.

Example 9 — Application — cost from a triangular area

A triangular garden has sides 18 m and 24 m with an included angle of 55ř. Sod costs \$8 per m^2. Find the total cost.

Solution. Area $= \frac{1}{2}(18)(24)\sin 55$ř ≈ 176.9 m^2.

Conclusion: cost $\approx 176.9 \times 8 \approx \$1415.20$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

In $\triangle ABC$, $A = 55$ř, $C = 65$ř, $c = 18$. Find a.

Question 2

A triangle has sides $7, 9, 13$. Find the largest angle.

Question 3 — Challenge

In $\triangle ABC$, $A = 28$ř, $a = 8$, $b = 15$. Determine whether the ambiguous case applies, and find every valid value of B.

Question 4

Find the area of a triangle with sides 9 and 12 and included angle 65ř.

Question 5

A hiker walks 4 km on a bearing of 030ř, then 6 km on a bearing of 120ř. Find the direct distance back to the start.

Question 6

In $\triangle ABC$, $a = 10$, $b = 14$, $C = 50$ř. Find c.

Question 7

A triangular field has sides 80 m, 95 m, and an included angle of 72ř between them. Find its area in m^2.

Question 8

In $\triangle ABC$, $b = 9$, $c = 13$, $A = 100$ř. Find a.

Question 9

In $\triangle ABC$, $A = 125$ř, $a = 18$, $b = 10$. Find B, explaining why only one triangle is possible.

Question 10

Find the area of a triangle with sides $7, 9, 11$ (find the angle first).

Question 11 — Challenge

In $\triangle ABC$, $A = 48$ř, $a = 11$, $c = 15$. Find angle C, and comment on whether the ambiguous case applies.

Question 12 — Challenge

A ship sails 30 km on bearing 025ř, then 45 km on bearing 110ř. Find the direct distance between the start and end points.

Question 13 — Challenge

A triangular plot has sides 20 m and 28 m with an included angle of 48ř; sod costs \$6/m^2. Find the total cost.

Question Recap & Answer Key

Question Recap

1. In $\triangle ABC$, $A = 55\text{ř}$, $C = 65\text{ř}$, $c = 18$. Find a.
2. A triangle has sides $7, 9, 13$. Find the largest angle.
3. In $\triangle ABC$, $A = 28\text{ř}$, $a = 8$, $b = 15$. Determine whether the ambiguous case applies, and find every valid value of B.
4. Find the area of a triangle with sides 9 and 12 and included angle 65ř.
5. A hiker walks 4 km on a bearing of 030ř, then 6 km on a bearing of 120ř. Find the direct distance back to the start.
6. In $\triangle ABC$, $a = 10$, $b = 14$, $C = 50\text{ř}$. Find c.
7. A triangular field has sides 80 m, 95 m, and an included angle of 72ř between them. Find its area in m^2.
8. In $\triangle ABC$, $b = 9$, $c = 13$, $A = 100\text{ř}$. Find a.
9. In $\triangle ABC$, $A = 125\text{ř}$, $a = 18$, $b = 10$. Find B, explaining why only one triangle is possible.
10. Find the area of a triangle with sides $7, 9, 11$ (find the angle first).
11. In $\triangle ABC$, $A = 48\text{ř}$, $a = 11$, $c = 15$. Find angle C, and comment on whether the ambiguous case applies.
12. A ship sails 30 km on bearing 025ř, then 45 km on bearing 110ř. Find the direct distance between the start and end points.
13. A triangular plot has sides 20 m and 28 m with an included angle of 48ř; sod costs $\$6/\text{m}^2$. Find the total cost.

✓ Answer Key

1. $a = \dfrac{18\sin 55\text{ř}}{\sin 65\text{ř}} \approx 16.3$
2. $\cos C = \dfrac{49 + 81 - 169}{126} = \dfrac{-39}{126} \Rightarrow C \approx 107.9\text{ř}$
3. $\sin B = \dfrac{15\sin 28\text{ř}}{8} \approx 0.880$; $B_1 \approx 61.6\text{ř}$, $B_2 = 118.4\text{ř}$; check $A + B_2 = 146.4\text{ř} < 180\text{ř}$: both valid
4. $\frac{1}{2}(9)(12)\sin 65\text{ř} \approx 48.98$
5. included angle $= 120 - 30 = 90\text{ř}$: $d^2 = 16 + 36 = 52 \Rightarrow d \approx 7.21$ km
6. $c^2 = 100 + 196 - 280\cos 50\text{ř} \approx 296 - 180.0 = 116.0 \Rightarrow c \approx 10.77$
7. $\frac{1}{2}(80)(95)\sin 72\text{ř} \approx 3616.6\ \text{m}^2$
8. $a^2 = 81 + 169 - 2(9)(13)\cos 100\text{ř} \approx 250 + 40.6 = 290.6 \Rightarrow a \approx 17.05$
9. $\sin B = \frac{10\sin 125\text{ř}}{18} \approx 0.4551 \Rightarrow B \approx 27.1\text{ř}$; the supplementary option would make $A + B_2 > 180\text{ř}$, so only one triangle exists
10. $\cos C = \frac{49+81-121}{126} \approx 0.0714 \Rightarrow C \approx 85.9\text{ř}$; Area $= \frac{1}{2}(7)(9)\sin 85.9\text{ř} \approx 31.4$
11. $\sin C = \dfrac{15\sin 48\text{ř}}{11} \approx 1.013$ — impossible (sin can't exceed 1), so no triangle exists with these values; the ambiguous case doesn't apply because there's no valid triangle at all
12. included angle $= 110 - 25 = 85\text{ř}$; $d^2 = 900 + 2025 - 2700\cos 85\text{ř} \approx 2925 - 235.4 = 2689.6 \Rightarrow d \approx 51.86$ km
13. Area $= \frac{1}{2}(20)(28)\sin 48\text{ř} \approx 208.1\ \text{m}^2$; cost $\approx 208.1 \times 6 \approx \$1248.60$